Department of Statistics & Mathematical Sciences

Problem Set 1: Probability Spaces, Central Limit Theorem, and Contingency Inference

Due: Friday, October 17 at 17:00 • Total Score: 100 Points

Problem 1: Moment Generating Functions and the Central Limit Theorem (50 Points)

Let $X_1, X_2, \ldots, X_n$ be a sequence of independent and identically distributed (i.i.d.) random variables with mean $\mu = 0$, variance $\sigma^2 = 1$, and moment generating function $M_X(t) = \mathbb{E}[e^{tX}]$ defined in a neighborhood of $t = 0$.

(a) Write the Taylor expansion of $M_X(t)$ around $t = 0$ up to second order.

(b) Define the standardized sum $Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} X_i$. Find the MGF $M_{Z_n}(t)$ in terms of $M_X(t)$.

(c) Prove that $\lim_{n\to\infty} M_{Z_n}(t) = e^{t^2/2}$, thereby establishing the Central Limit Theorem.

Detailed Solution & Mathematical Proof

Part (a): Expanding $M_X(t) = \mathbb{E}[1 + tX + \frac{t^2X^2}{2} + o(t^2)]$: Since $\mathbb{E}[X] = 0$ and $\mathbb{E}[X^2] = 1$, we obtain:

$$M_X(t) = 1 + \frac{t^2}{2} + o(t^2)$$

Part (b): By independence of X_i:

$$M_{Z_n}(t) = \mathbb{E}\left[\exp\left(\frac{t}{\sqrt{n}} \sum_{i=1}^{n} X_i\right)\right] = \prod_{i=1}^{n} M_X\left(\frac{t}{\sqrt{n}}\right) = \left[M_X\left(\frac{t}{\sqrt{n}}\right)\right]^n$$

Part (c): Substituting the Taylor expansion:

$$M_{Z_n}(t) = \left[1 + \frac{t^2}{2n} + o\left(\frac{t^2}{n}\right)\right]^n$$

Using the fundamental limit $\lim_{n\to\infty}(1 + \frac{a}{n})^n = e^a$, we set $a = t^2/2$:

$$\lim_{n\to\infty} M_{Z_n}(t) = \exp\left(\frac{t^2}{2}\right)$$

Since $e^{t^2/2}$ is the moment generating function of a standard normal distribution $\mathcal{N}(0,1)$, by the Continuity Theorem for MGFs, $Z_n \xrightarrow{d} \mathcal{N}(0,1)$. ■

Problem 2: Asymptotic Variance of the Log Odds Ratio (50 Points)

For a 2×2 contingency table with independent multinomial counts $(n_{11}, n_{12}, n_{21}, n_{22})$, the sample Odds Ratio is $\widehat{\text{OR}} = \frac{n_{11}n_{22}}{n_{12}n_{21}}$.

(a) Apply the Delta Method to derive the asymptotic variance of $\ln \widehat{\text{OR}}$.

(b) Derive the formula for a $(1-\alpha)$ confidence interval for the true population Odds Ratio.

Detailed Solution & Mathematical Proof

Part (a): Let $\theta = \ln \widehat{\text{OR}} = \ln n_{11} - \ln n_{12} - \ln n_{21} + \ln n_{22}$. Using the delta method for independent Poisson/multinomial components:

$$\text{Var}(\ln \widehat{\text{OR}}) \approx \sum_{i,j} \left(\frac{\partial \theta}{\partial n_{ij}} \right)^2 \text{Var}(n_{ij}) = \frac{1}{n_{11}} + \frac{1}{n_{12}} + \frac{1}{n_{21}} + \frac{1}{n_{22}}$$

Part (b): The standard error is $\text{SE}(\ln \widehat{\text{OR}}) = \sqrt{\frac{1}{n_{11}} + \frac{1}{n_{12}} + \frac{1}{n_{21}} + \frac{1}{n_{22}}}$. The $(1-\alpha)$ confidence interval for OR is:

$$\left[\exp\left(\ln \widehat{\text{OR}} - z_{1-\alpha/2}\text{SE} \right), \ \exp\left(\ln \widehat{\text{OR}} + z_{1-\alpha/2}\text{SE} \right) \right] \quad \blacksquare$$